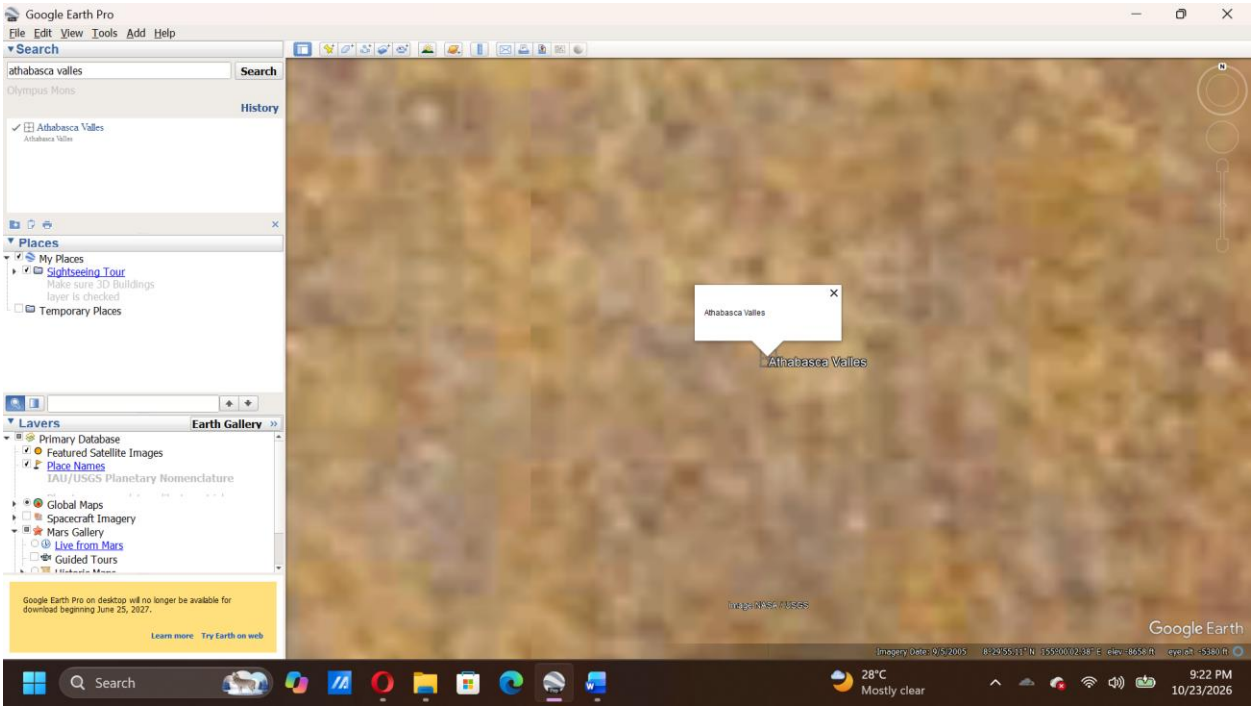
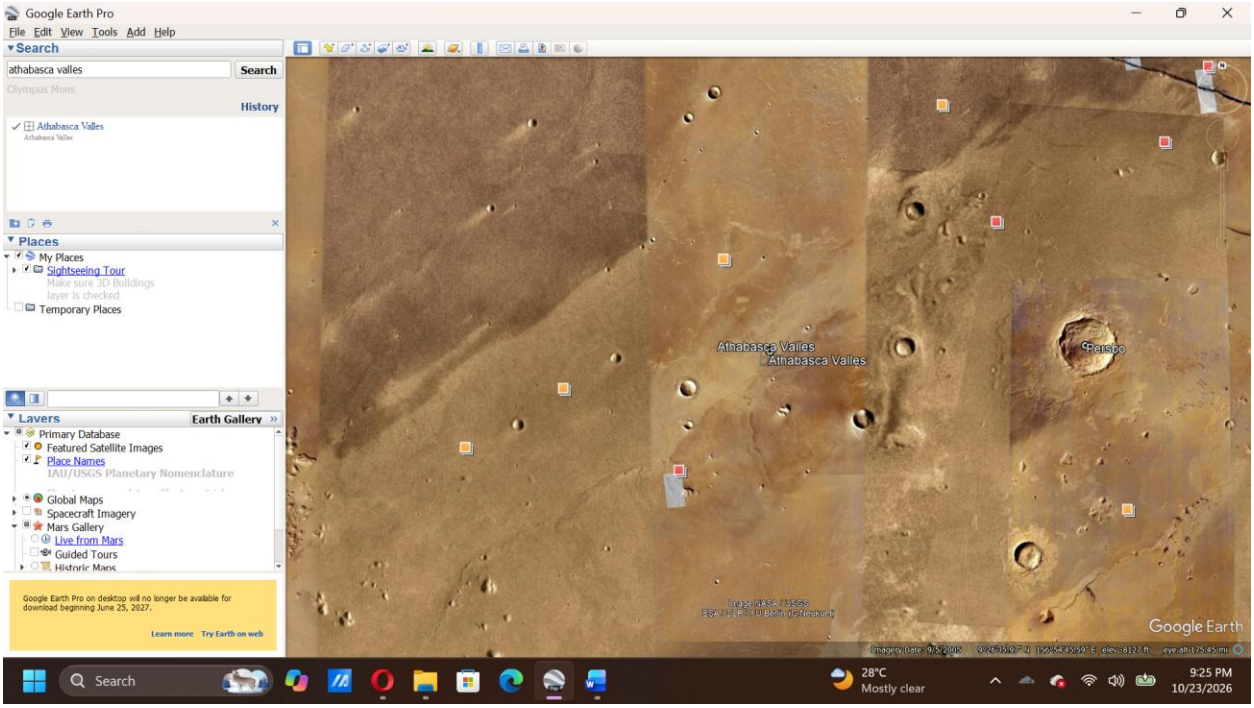
Task 1

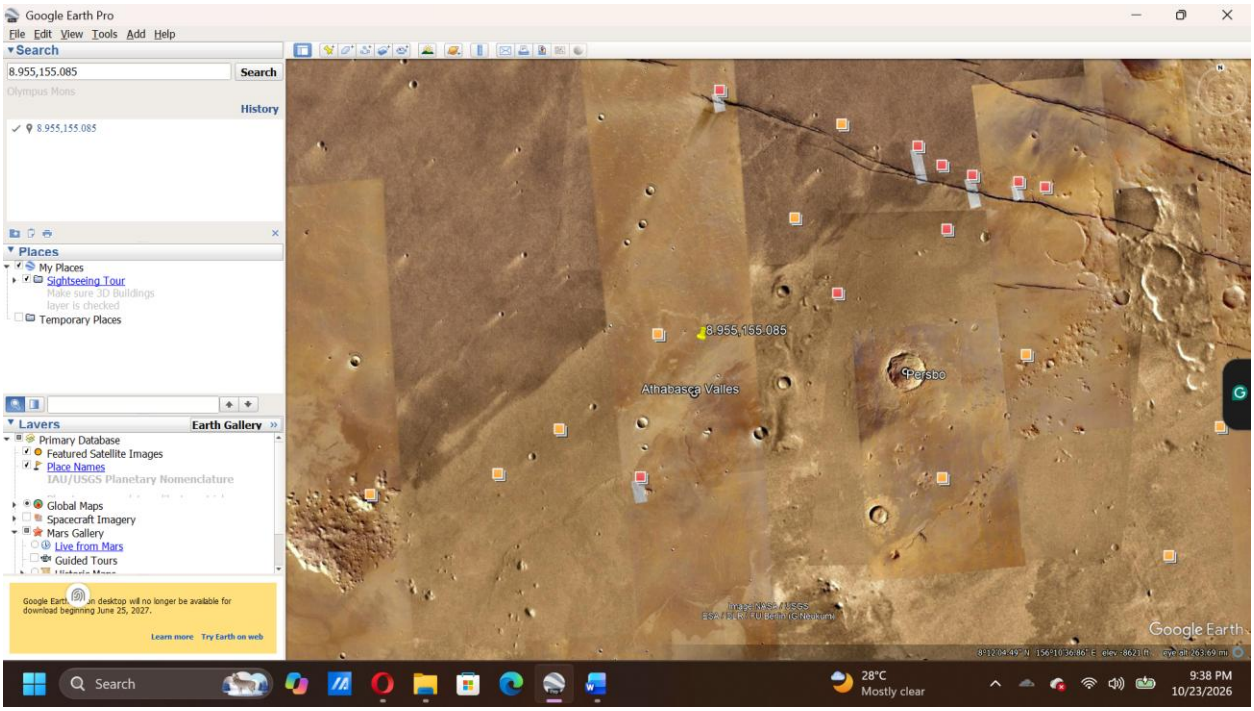
While selecting a landing site, the first thing we should keep in mind is how flat the surface is because the safe landing of our rover matters the most. The second most important thing is to search for a place with the least amount of obstacles to ensure that our rover remains unaffected by its rocks, craters and cracks. Then comes the third part which directly correlates to our scientific investigation: the geological features but at the same time we also have to ensure that the place is accessible and our rover will not leave us halfway through. Lastly, the place must also have some sort of scientific importance too which enables us to learn more about the geological history of the planet.

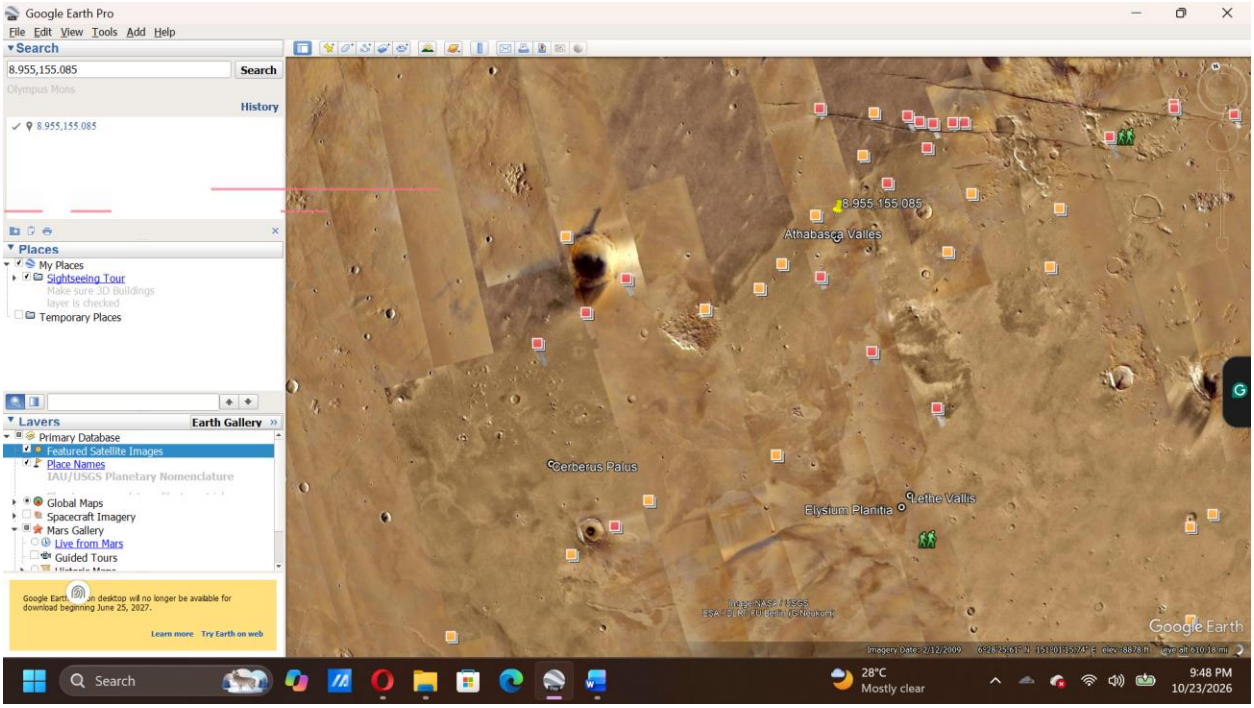


This is the initial image I got from Google Earth Pro after searching for Athabasca Valles.



This is the zoomed out version which has a clearer view but clearly shows a lot of bumps but right above the Athabasca Valles writing, I noticed several plain land type area with fewer bumps and the image also had yellow and red colored square-shapes further inspection of which led me select a place closer to these because it can give clearer HiRISE image providing detailed view and hence easing the process of site selection.

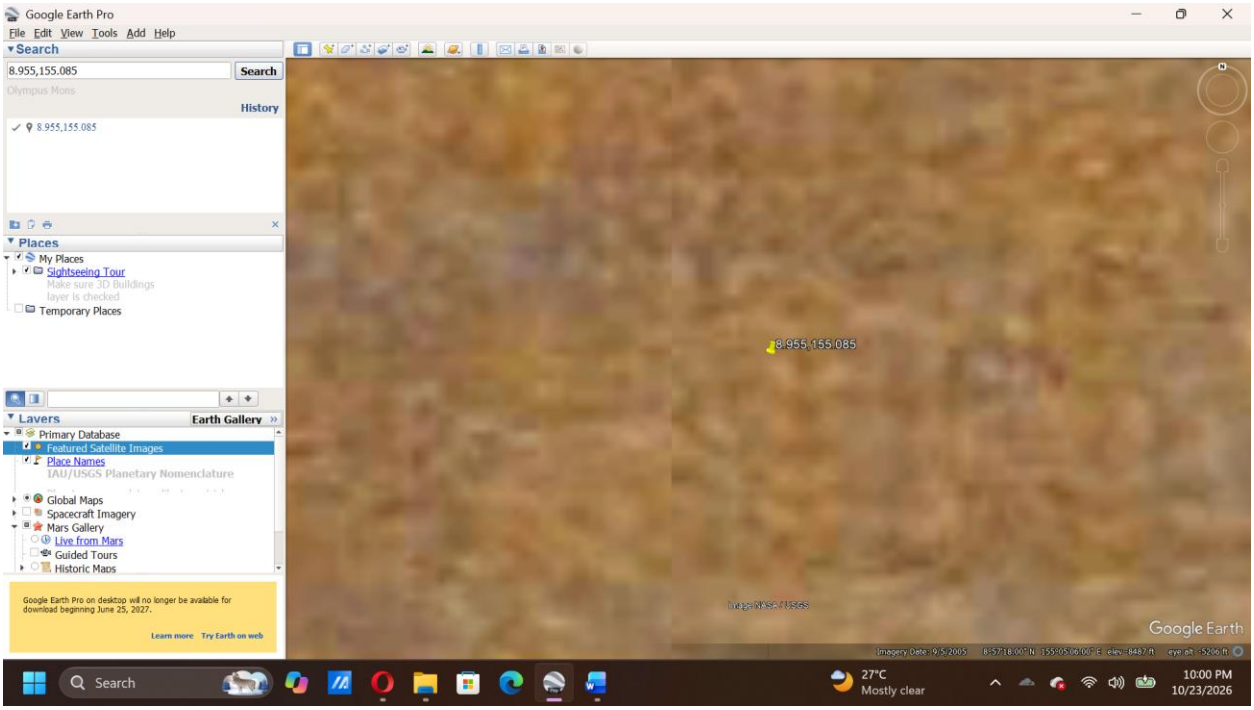


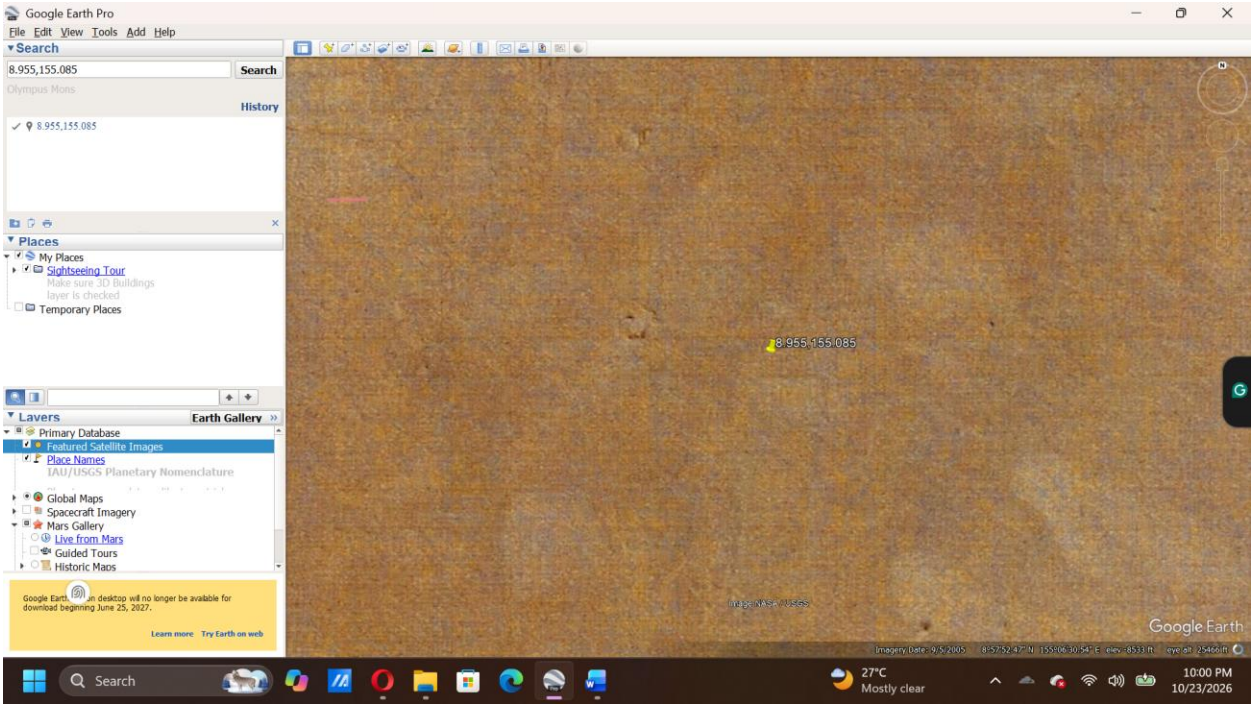


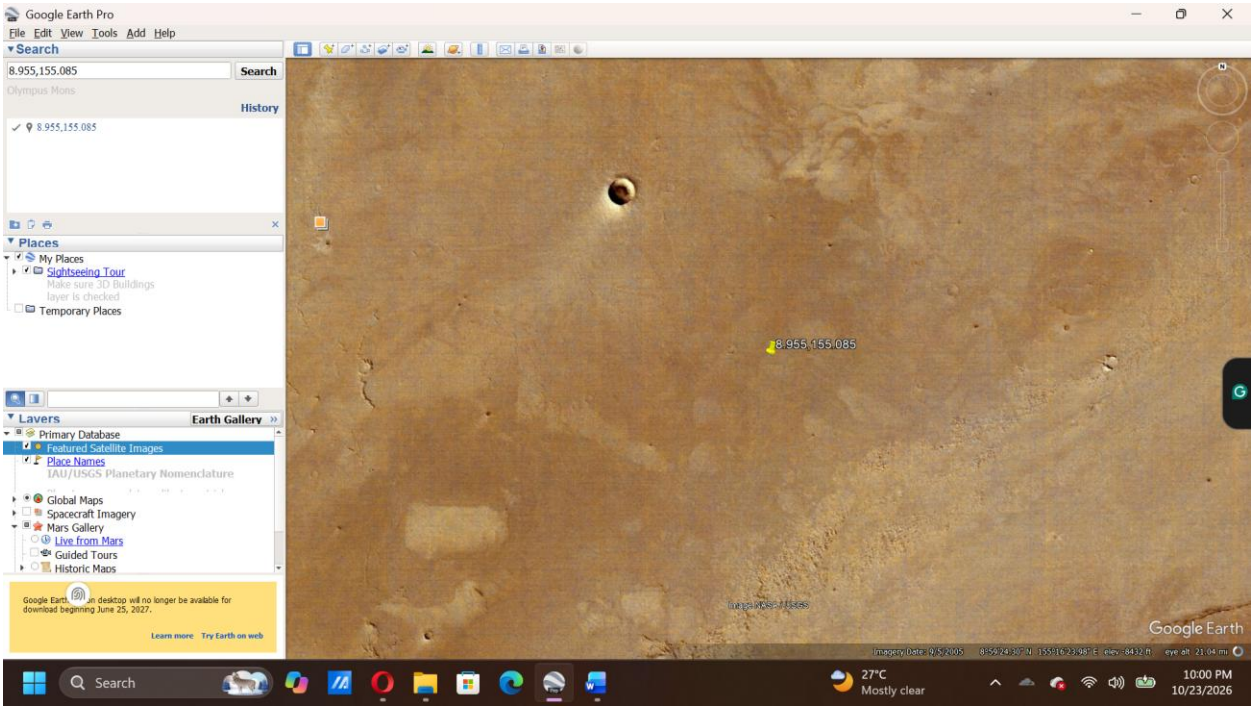
Finally, I have selected the yellow marked area from the image because it showed the most square shape and a promising flat terrain.

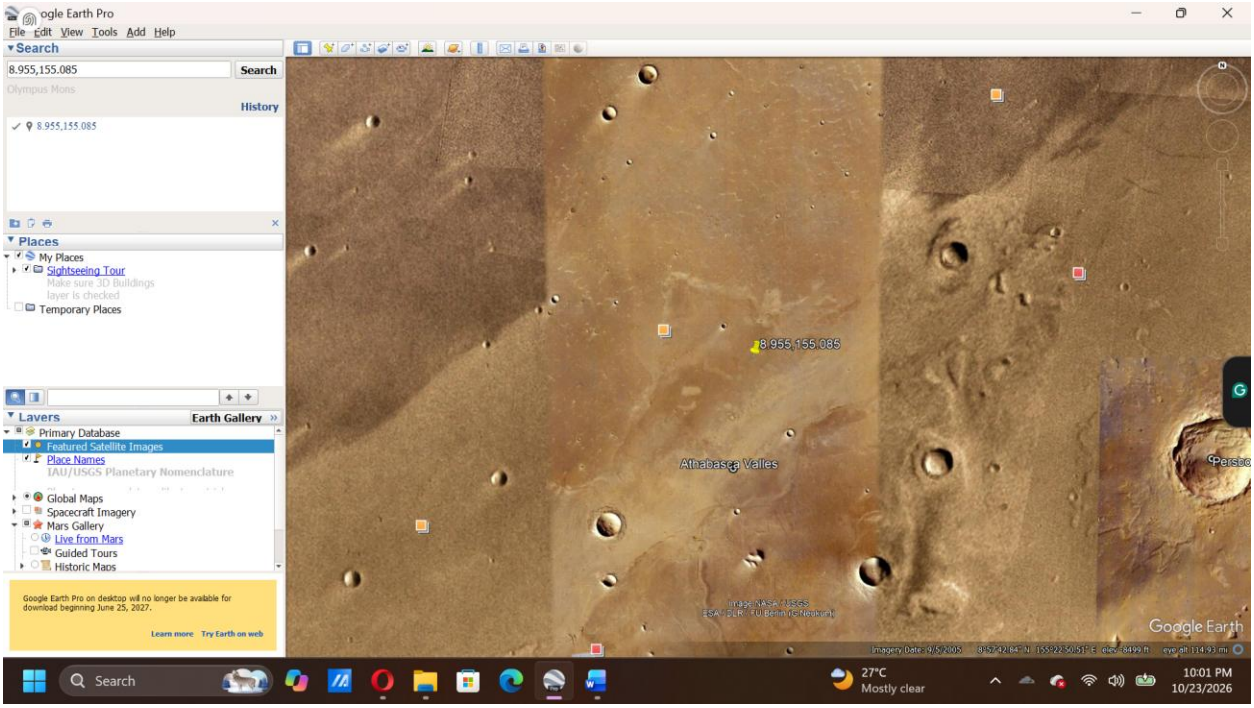
1. Landing Site Identification: I have finalized Central Athabasca Valles Lava Plain as our proposed landing-site area.

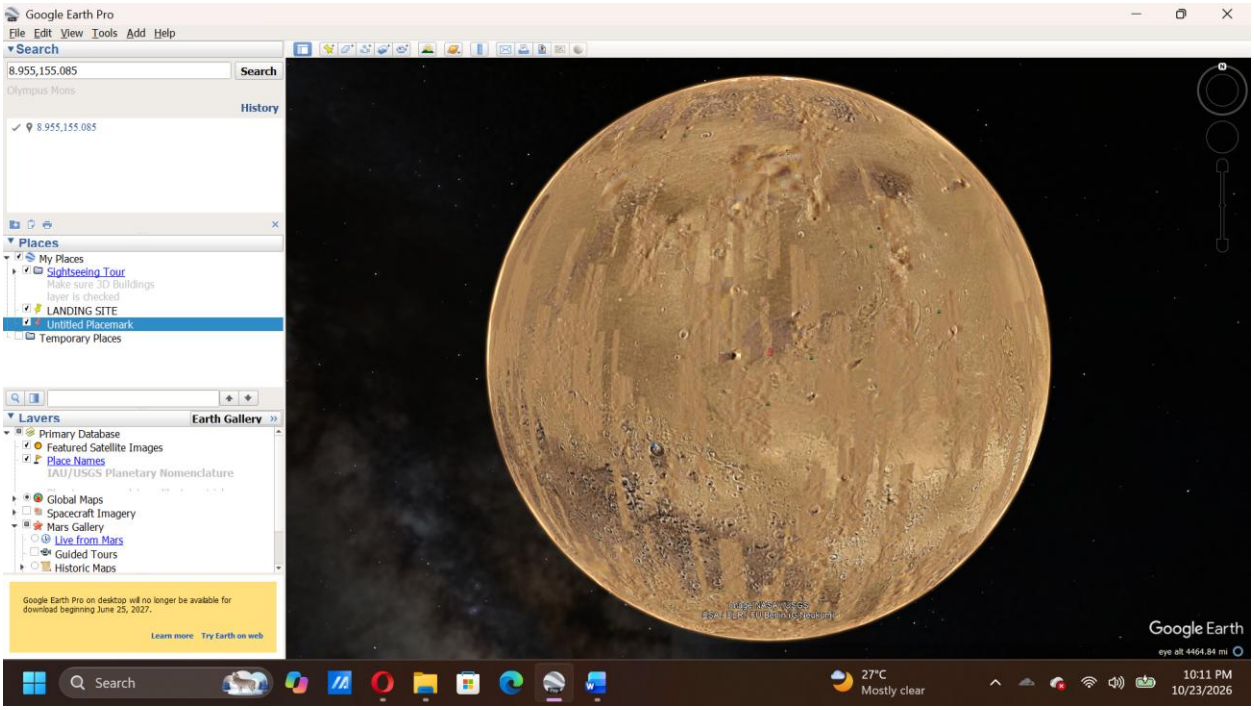
2. Global View:




Google Earth Pro
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Earth Gallery
Primary Database
Featured Satellite Images
Place Names
IAU/USGS Planetary Nomenclature
Global Maps
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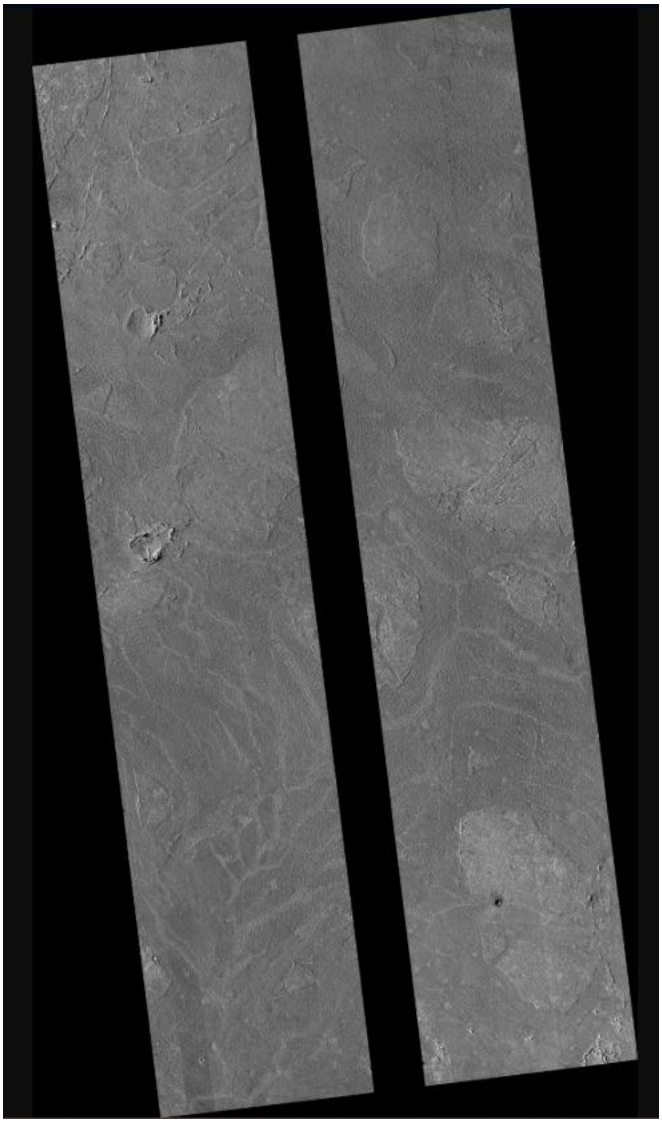

Google Earth Pro
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Place Names
IAU/USGS Planetary Nomenclature
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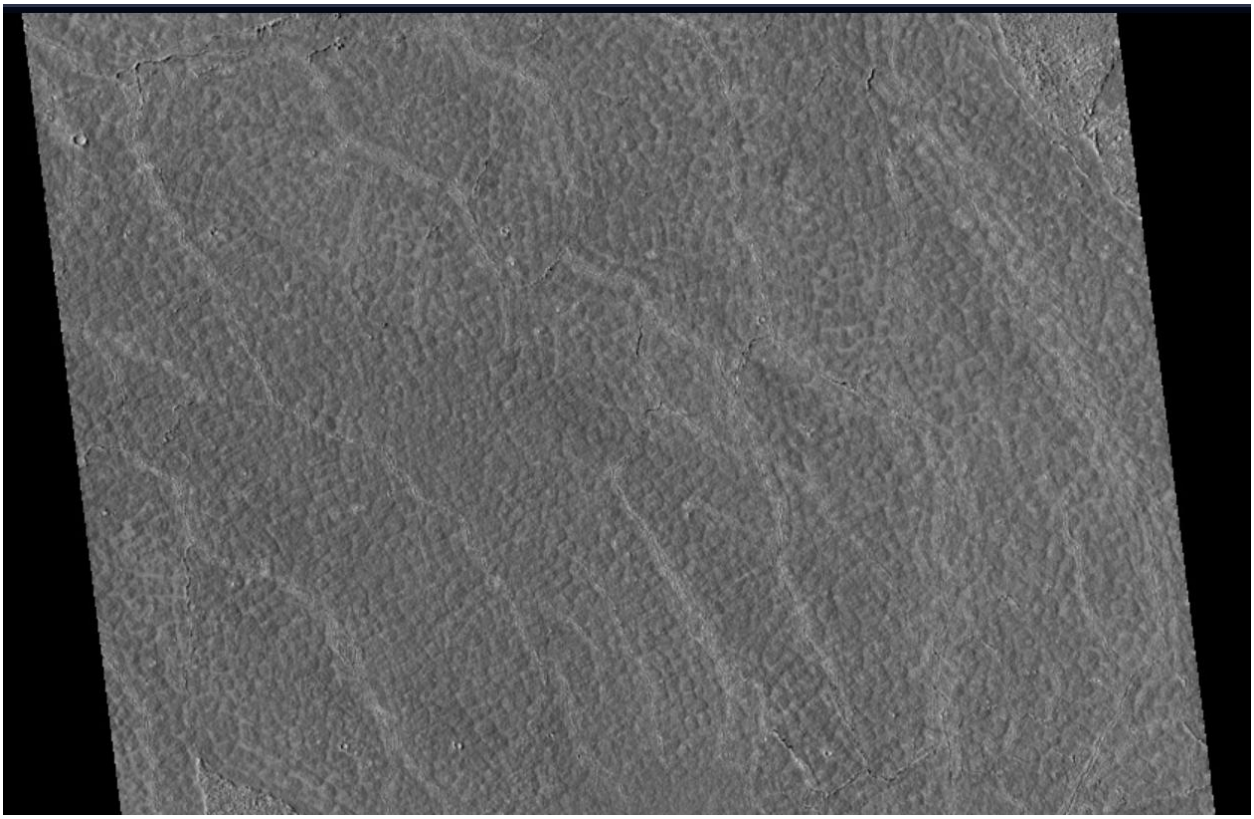




The red marked area is the selected site.

3. Close-up View:





This is a close up I have selected for landing from the middle left section.

4. Scientific Justification: The proposed landing-site area is located in Athabasca Valles, near 8.955° N, 155.085° E. The HiRISE imagery shows relatively uniform terrain with patterned surfaces and visible fractures. The area was selected because some portions appear less obstructed by large craters, while nearby geological features offer opportunities for rover exploration. Athabasca Valles is scientifically important because of evidence of past flooding and volcanic activity. Studying the surface textures and fractures could help us understand the region's volcanic history and geological evolution. However, further terrain analysis would be required to confirm landing safety.

TASK 2

Selected Landing-Site Area: Athabasca Valles, Mars
Reference Coordinates: 8.955° N, 155.085° E (HiRISE image center)

1. Primary Scientific Hypothesis

Athabasca Valles was shaped by large floods in the past, followed by volcanic activity that covered parts of the surface with lava.

Scientific Question: How did past flooding and volcanic activity contribute to the formation of Athabasca Valles?

2. Mini-Hypotheses

Mini-Hypothesis 1: Volcanic Activity

The rough patterns and cracks visible in the HiRISE images may have formed when lava cooled and hardened.

Scientific Question: Were these surface patterns formed by volcanic activity?

Samples: Basaltic rocks and rocks near fractured surfaces.

Reason: Basalt is a common volcanic rock. Examining its texture and composition could help us understand whether the surface was formed by lava.

Mini-Hypothesis 2: Past Water Activity

Water may have flowed through Athabasca Valles before the region was covered by lava, leaving behind sediments and erosion features.

Scientific Question: Can we find evidence of past water activity in the rocks or soil?

Samples: Fine-grained soil, sand, and possible sedimentary rocks.

Reason: Flowing water can transport and deposit sediments. Studying their size, shape, and arrangement might provide clues about past water movement.

Mini-Hypothesis 3: Possibility of Ancient Life

If water once existed in Athabasca Valles, some sediments or minerals might have preserved traces of past habitable conditions or possible microbial life.

Scientific Question: Are there any materials that could preserve possible signs of ancient life?

Samples: Fine-grained sedimentary rocks and clay-bearing materials, if available.

Reason: These materials can sometimes preserve organic compounds or chemical evidence related to past environments. However, finding such materials would not necessarily mean that life existed there.

3. Rover Investigation Plan

The rover will first examine the surrounding terrain using its cameras to identify interesting geological features and avoid obstacles.

Step	Investigation	Purpose
1	Observe the surrounding terrain	Identify rocks, cracks, and possible sediment deposits
2	Examine or collect a volcanic rock sample	Look for evidence of past lava flows
3	Collect a loose soil or sand sample	Study the characteristics of surface materials
4	Examine possible sedimentary rocks or clay-bearing materials	Look for possible evidence of past water activity
5	Compare the observations and samples	Evaluate the proposed hypotheses

The rover will photograph the selected samples and collect them if its sampling mechanism allows it. If suitable instruments are available, further examination of the samples could help identify their mineral composition.

4. Expected Outcome

Through this investigation, I hope to understand how flooding and volcanic activity shaped Athabasca Valles. The rock and soil samples may provide clues about the region's geological history and whether it once had conditions suitable for life.

Since this is a simulated rover mission, the investigation will focus on observing geological features and collecting suitable samples rather than confirming the presence of ancient life.

Reference

University of Arizona, HiRISE Observation ESP_090036_1890, *Athabasca Valles Lava*. https://www.uahirise.org/ESP_090036_1890

TASK 3

1. Working Principle

Raman spectroscopy is a technique used to identify minerals by shining a laser onto a sample and analyzing the scattered light. A small portion of this light changes wavelength due to interactions with the material, producing a Raman spectrum that acts like a fingerprint for identifying minerals.

I propose using a **532 nm Raman spectrometer** to examine rock and soil samples from simulated Martian terrain.

2. Required Components and Estimated Prices

For this design, I considered a low-cost Raman spectrometer inspired by the open-source DIYraman project.

Component	Purpose	Estimated Cost (BDT)
Used B&W Tek spectrometer with detector	Records Raman spectra	৳25,000
532 nm laser	Illuminates the sample	৳1,875
550 nm longpass filter	Blocks unwanted laser light	৳21,250
550 nm dichroic mirror	Directs the laser and separates returning light	৳28,750
532 nm bandpass filter	Cleans the laser beam	৳11,875
Two adjustable optical mounts	Holds and aligns components	৳11,250
Microscope objective	Focuses light onto the sample	৳11,250
Quartz cuvette/sample holder	Holds samples for examination	৳2,500
Optical fiber	Transfers light to the spectrometer	৳6,250
3D-printing materials	Supports the optical assembly	৳3,125
Subtotal		**৳1,23,125**

Prices are converted from the DIYraman reference component list using an assumed exchange rate of $1 = ৳125. They are not confirmed quotations from Bangladesh suppliers.

Component and Supplier Links:

1. DIYraman – Original design and component prices
2. DIYraman – Detailed Bill of Materials

3. Thorlabs – Optical filters, mirrors and mounts
4. Edmund Optics – 532 nm optical filters
5. eBay – Used B&W Tek spectrometer
6. AliExpress – Laser modules and optical fibers

Most specialized optical components would need to be imported, while the enclosure and some mounting parts could be fabricated locally in Bangladesh.

3. Optical Layout and Rover Integration

The Raman spectrometer would be placed inside the rover's science payload box.

The optical arrangement would follow this path:

532 nm Laser → Bandpass Filter → Dichroic Mirror → Microscope Objective → Rock/Soil Sample

Scattered Light → Microscope Objective → Dichroic Mirror → Longpass Filter → Spectrometer → Computer

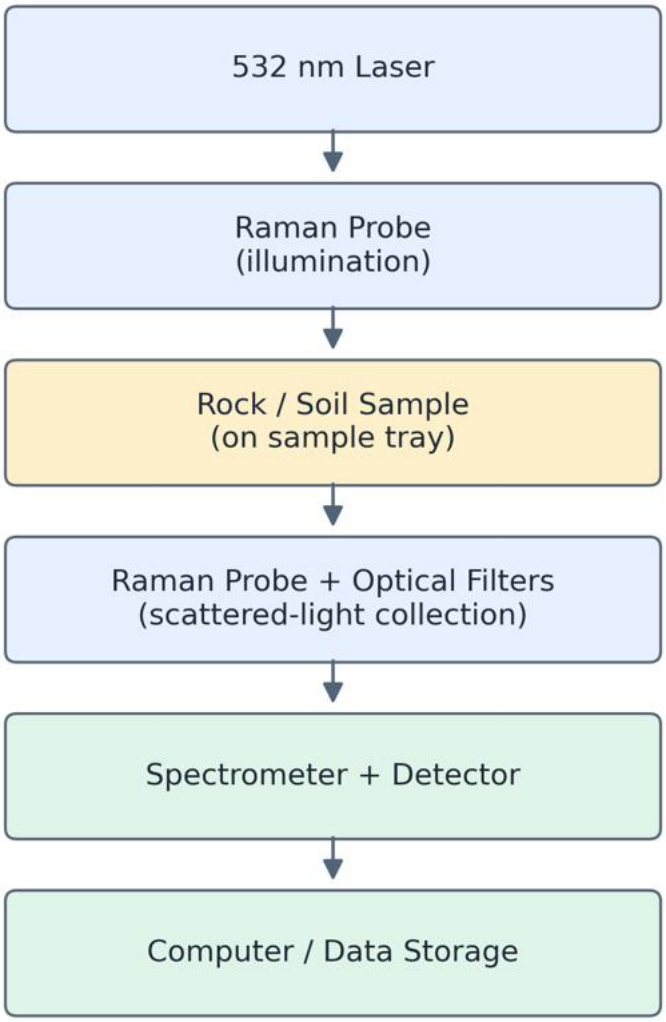


Figure 1: Conceptual optical layout of the Raman spectrometer inside the rover's science payload box.

The sample would be placed on a small tray beneath the microscope objective. The laser would illuminate it, and the scattered light would be collected and directed toward the spectrometer.

The components would be mounted inside a light-tight enclosure to protect them from dust, vibrations and accidental laser exposure.

4. Application to Task 2 Hypotheses

The instrument could help investigate the geological hypotheses proposed for Athabasca Valles.

Hypothesis	Samples	Purpose

Volcanic activity	Basaltic rocks	Identify minerals such as olivine and pyroxene
Past water activity	Sediments and altered rocks	Search for minerals such as gypsum, carbonates or clay minerals
Possible ancient life	Fine-grained sediments	Investigate possible organic compounds or minerals associated with past habitable conditions

Raman spectroscopy could help identify the mineral composition of samples and provide clues about the geological history of Athabasca Valles.

However, detecting organic compounds would not directly confirm the existence of ancient life. Some minerals and organic signals may also be difficult to detect with this low-cost setup.

5. Design Justification and Limitations

I selected a 532 nm laser because it is relatively affordable and can produce strong Raman signals. Using a second-hand spectrometer and 3D-printed components would also help reduce the cost.

The main limitations of this design are:

- **Fluorescence:** Some minerals or organic materials may produce fluorescence that interferes with Raman signals.
- **Weak signals:** A sensitive detector and accurate optical alignment would be necessary.
- **Limited spectral range:** The proposed 550 nm longpass filter would prevent measurement of some lower Raman shifts.
- **Dust and vibration:** These could affect optical alignment and measurement quality.
- **Laser safety:** Protective equipment, an enclosed optical path and a safety interlock would be necessary.

The instrument would also need calibration using known mineral samples before testing unknown materials.

Since this is a conceptual design, the final dimensions, weight, power consumption and performance would need to be checked before integrating it into the rover.

6. Estimated Total Budget

Along with the main optical components, I included additional allowances for safety equipment, electronics and importing components to Bangladesh.

Budget Category	Estimated Cost (BDT)
Main Raman components	৳1,23,125
Laser safety equipment and enclosure	৳20,000
Controller, power supply and electronics	৳8,000
Shipping and customs allowance	৳12,000
Subtotal	**৳1,63,125**
Contingency (20%)	৳32,625
Total Estimated Budget	**৳1,95,750**

A simpler laboratory prototype could be developed first, with rover integration considered after successful testing.

7. Conclusion

The proposed Raman spectrometer could help the rover examine mineral samples related to volcanic activity and past water activity in Athabasca Valles.

By using second-hand optical equipment and locally fabricated parts, the overall cost could be reduced. Although the design has technical limitations, I believe it could be a useful starting point for developing a Raman-based science instrument for a student rover team.

8. References

1. DIYraman. *DIY Raman Spectroscopy and Component Prices*. https://diyraman.com/
2. DIYraman Documentation. *Detailed Bill of Materials*. https://docs.diyraman.com/bom/BOM/
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4. B&W Tek. *Discover-It-Yourself Raman Kits*. https://bwtek.com/products/diy-system/diy_raman_bwtek/
5. Thorlabs. *Optical Components and Filters*. https://www.thorlabs.com/
6. Edmund Optics. *Optical Filters*. https://www.edmundoptics.com/
7. NASA Jet Propulsion Laboratory. *SHERLOC: Looking for Life and Minerals with UV Light*. https://microdevices.jpl.nasa.gov/capabilities/optical-components/sherloc/